ASSIGNMENT – IX

TITLE :
Implement auto-scaling for your AI app on Kubernetes and monitor its health and performance metrics.

NAME : Shinde Shubham Dnyandev. **ROLL NO :** 23107121. **BATCH :** B

```
admin@ubantu-Precision-3660: ~/ai-k8s-app
admin@ubantu-Precision-3660: ~/ai-k8s-app   admin@ubantu-Precision-3660: ~/ai-k8s-app   admin@ubantu-Precision-3660: ~/ai-k8s-app   admin@ubantu-Precision-3660: ~/ai-k8s-app
admin@ubantu-Precision-3660:~$ sudo systemctl start docker
admin@ubantu-Precision-3660:~$ sudo systemctl enable docker
Synchronizing state of docker.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable docker
admin@ubantu-Precision-3660:~$ docker --version
Docker version 29.3.0, build 5927d80
admin@ubantu-Precision-3660:~$ kubectl version --client
Client Version: v1.35.2
Kustomize Version: v5.7.1
admin@ubantu-Precision-3660:~$ minikube version
minikube version: v1.38.1
commit: c93a4cb9311efc66b90d33ea03f75f2c4120e9b0
admin@ubantu-Precision-3660:~$ minikube start --driver=docker
😄  minikube v1.38.1 on Ubuntu 24.04
✨  Using the docker driver based on existing profile
👍  Starting "minikube" primary control-plane node in "minikube" cluster
🚜  Pulling base image v0.0.50 ...
🏃  Updating the running docker "minikube" container ...
🐳  Preparing Kubernetes v1.35.1 on Docker 29.2.1 ...
🔎  Verifying Kubernetes components...
    ▪ Using image registry.k8s.io/metrics-server/metrics-server:v0.8.1
    ▪ Using image gcr.io/k8s-minikube/storage-provisioner:v5
🌟  Enabled addons: default-storageclass, storage-provisioner, metrics-server
🏄  Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default
admin@ubantu-Precision-3660:~$ kubectl get nodes
NAME       STATUS   ROLES           AGE     VERSION
minikube   Ready    control-plane   3m55s   v1.35.1
admin@ubantu-Precision-3660:~$ mkdir ai-k8s-app
admin@ubantu-Precision-3660:~$ cd ai-k8s-app
admin@ubantu-Precision-3660:~/ai-k8s-app$ nano app.py
```

```
admin@ubantu-Precision-3660: ~/ai-k8s-app
  GNU nano 7.2                                                    app.py *
from flask import Flask, request, jsonify

app = Flask(__name__)

@app.route("/")
def home():
    return "AI Model Running in Kubernetes"

@app.route("/predict", methods=["POST"])
def predict():
    data = request.json["number"]
    result = data * 2
    return jsonify({"result": result})

if __name__ == "__main__":
    app.run(host="0.0.0.0", port=5000)
```

```
admin@ubantu-Precision-3660:~/ai-k8s-app$ nano requirements.txt
```

```
admin@ubantu-Precision-3660: ~/ai-k8s-app
  GNU nano 7.2                                                    requirements.txt *
Flask
```

```
admin@ubantu-Precision-3660:~/ai-k8s-app$ nano Dockerfile
```

admin@ubantu-Precision-3660: ~/ai-k8s-app

```
  GNU nano 7.2                                                    Dockerfile *
FROM python:3.9
WORKDIR /app
COPY . .
RUN pip install flask
EXPOSE 5000
CMD ["python", "app.py"]
```

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```
admin@ubantu-Precision-3660:~/ai-k8s-app$ sudo docker build -t ai-model .
[+] Building 32.5s (9/9) FINISHED                                                                                    docker:default
 => [internal] load build definition from Dockerfile                                                                                0.0s
 => => transferring dockerfile: 134B                                                                                                0.0s
 => [internal] load metadata for docker.io/library/python:3.9                                                                       2.0s
 => [internal] load .dockerignore                                                                                                   0.0s
 => => transferring context: 2B                                                                                                     0.0s
 => [1/4] FROM docker.io/library/python:3.9@sha256:da5aee29682d12a6649f51c8d6f15b87deb3e6c524b923c41d0cb3304d07c913                 0.0s
 => [internal] load build context                                                                                                   0.0s
 => => transferring context: 579B                                                                                                   0.0s
 => CACHED [2/4] WORKDIR /app                                                                                                       0.0s
 => [3/4] COPY . .                                                                                                                  0.0s
 => [4/4] RUN pip install flask                                                                                                    30.4s
 => exporting to image                                                                                                              0.1s
 => => exporting layers                                                                                                             0.1s
 => => writing image sha256:0e4f7d42ebf6a4ee0c8ec9dd7f041255c4f4073494d74957a7279bddc477cecc                                        0.0s
 => => naming to docker.io/library/ai-model                                                                                         0.0s
admin@ubantu-Precision-3660:~/ai-k8s-app$ sudo docker images
                                                                                                                  i Info →  U  In Use
IMAGE                                      ID             DISK USAGE   CONTENT SIZE   EXTRA
ai-docker-app-frontend:latest              dd4316eaf002       62.1MB             0B
ai-docker-app-model:latest                 72b751b8dcc7       1.11GB             0B
ai-model:latest                            0e4f7d42ebf6        1.1GB             0B
gcr.io/k8s-minikube/kicbase:v0.0.50        6da180ef5035       1.37GB             0B    U
hello-world:latest                         1b44b5a3e06a       10.1kB             0B    U
mongo:latest                               f0286cc18189        942MB             0B
admin@ubantu-Precision-3660:~/ai-k8s-app$ minikube image load ai-model
admin@ubantu-Precision-3660:~/ai-k8s-app$ nano deployment.yaml
```

admin@ubantu-Precision-3660: ~/ai-k8s-app

```
  GNU nano 7.2                                                    deployment.yaml *
apiVersion: apps/v1
kind: Deployment
metadata:
  name: ai-app
spec:
  replicas: 2
  selector:
    matchLabels:
      app: ai-app
  template:
    metadata:
      labels:
        app: ai-app
    spec:
      containers:
      - name: ai-container
        image: ai-model
        imagePullPolicy: Never
        ports:
        - containerPort: 5000
        resources:
          requests:
            cpu: "100m"
          limits:
            cpu: "500m"
```

```
admin@ubantu-Precision-3660:~/ai-k8s-app$ kubectl apply -f deployment.yaml
deployment.apps/ai-app created
admin@ubantu-Precision-3660:~/ai-k8s-app$ kubectl get pods
NAME                        READY   STATUS    RESTARTS   AGE
ai-app-5c86c57794-l8hkg     1/1     Running   0          6s
ai-app-5c86c57794-mb7rv     1/1     Running   0          6s
admin@ubantu-Precision-3660:~/ai-k8s-app$ nano service.yaml
```

admin@ubantu-Precision-3660: ~/ai-k8s-app

```
  GNU nano 7.2                                                    service.yaml *
apiVersion: v1
kind: Service
metadata:
  name: ai-service
spec:
  type: NodePort
  selector:
    app: ai-app
  ports:
  - port: 5000
    targetPort: 5000
    nodePort: 30007
```

```
admin@ubantu-Precision-3660:~/ai-k8s-app$ kubectl apply -f service.yaml
service/ai-service created
admin@ubantu-Precision-3660:~/ai-k8s-app$ kubectl get svc
NAME         TYPE       CLUSTER-IP     EXTERNAL-IP   PORT(S)          AGE
ai-service   NodePort   10.101.23.118  <none>        5000:30007/TCP   8s
kubernetes   ClusterIP  10.96.0.1      <none>        443/TCP          10m
admin@ubantu-Precision-3660:~/ai-k8s-app$ minikube service ai-service
```

NAMESPACE	NAME	TARGET PORT	URL
default	ai-service	5000	http://192.168.49.2:30007

```
🎉  Opening service default/ai-service in default browser...
admin@ubantu-Precision-3660:~/ai-k8s-app$ Gtk-Message: 16:28:09.404: Not loading module "atk-bridge": The functionality is provided by GTK natively. Please try to not load it.
[56226, Main Thread] WARNING: GTK+ module /snap/firefox/7901/gnome-platform/usr/lib/gtk-2.0/modules/libcanberra-gtk-module.so cannot be loaded.
GTK+ 2.x symbols detected. Using GTK+ 2.x and GTK+ 3 in the same process is not supported.: 'glib warning', file /build/firefox/parts/firefox/build/toolkit/xre/nsSigHandlers.cpp:201

(firefox_firefox:56226): Gtk-WARNING **: 16:28:09.445: GTK+ module /snap/firefox/7901/gnome-platform/usr/lib/gtk-2.0/modules/libcanberra-gtk-module.so cannot be loaded.
GTK+ 2.x symbols detected. Using GTK+ 2.x and GTK+ 3 in the same process is not supported.
Gtk-Message: 16:28:09.445: Failed to load module "canberra-gtk-module"
[56226, Main Thread] WARNING: GTK+ module /snap/firefox/7901/gnome-platform/usr/lib/gtk-2.0/modules/libcanberra-gtk-module.so cannot be loaded.
GTK+ 2.x symbols detected. Using GTK+ 2.x and GTK+ 3 in the same process is not supported.: 'glib warning', file /build/firefox/parts/firefox/build/toolkit/xre/nsSigHandlers.cpp:201

(firefox_firefox:56226): Gtk-WARNING **: 16:28:09.445: GTK+ module /snap/firefox/7901/gnome-platform/usr/lib/gtk-2.0/modules/libcanberra-gtk-module.so cannot be loaded.
GTK+ 2.x symbols detected. Using GTK+ 2.x and GTK+ 3 in the same process is not supported.
Gtk-Message: 16:28:09.445: Failed to load module "canberra-gtk-module"
admin@ubantu-Precision-3660:~/ai-k8s-app$
```

192.168.49.2:30007/

Not Secure http://192.168.49.2:30007

AI Model Running in Kubernetes

admin@ubantu-Precision-3660: ~/ai-k8s-app

```
admin@ubantu-Precision-3660:~/ai-k8s-app$ minikube addons enable metrics-server
💡  metrics-server is an addon maintained by Kubernetes. For any concerns contact minikube on GitHub.
You can view the list of minikube maintainers at: https://github.com/kubernetes/minikube/blob/master/OWNERS
    ▪ Using image registry.k8s.io/metrics-server/metrics-server:v0.8.1
🌟  The 'metrics-server' addon is enabled
admin@ubantu-Precision-3660:~/ai-k8s-app$ kubectl top pods
NAME                      CPU(cores)   MEMORY(bytes)
ai-app-5c86c57794-l8hkg   1m           19Mi
ai-app-5c86c57794-mb7rv   1m           19Mi
admin@ubantu-Precision-3660:~/ai-k8s-app$ kubectl autoscale deployment ai-app --cpu-percent=50 --min=2 --max=10
Flag --cpu-percent has been deprecated, Use --cpu with percentage or resource quantity format (e.g., '70%' for utilization or '500m' for milliCPU).
horizontalpodautoscaler.autoscaling/ai-app autoscaled
admin@ubantu-Precision-3660:~/ai-k8s-app$ kubectl get hpa
NAME     REFERENCE           TARGETS              MINPODS   MAXPODS   REPLICAS   AGE
ai-app   Deployment/ai-app   cpu: <unknown>/50%   2         10        0          5s
admin@ubantu-Precision-3660:~/ai-k8s-app$ while true; do curl $(minikube service ai-service --url); done
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```

admin@ubantu-Precision-3660: ~/ai-k8s-app

```
admin@ubantu-Precision-3660:~/ai-k8s-app$ kubectl get pods -w
NAME                      READY   STATUS    RESTARTS   AGE
ai-app-5c86c57794-l8hkg   1/1     Running   0          2m57s
ai-app-5c86c57794-mb7rv   1/1     Running   0          2m57s
```

```
admin@ubantu-Precision-3660:~/ai-k8s-app$ kubectl get pods
NAME                      READY   STATUS    RESTARTS   AGE
ai-app-5c86c57794-l8hkg   1/1     Running   0          7m42s
ai-app-5c86c57794-mb7rv   1/1     Running   0          7m42s
admin@ubantu-Precision-3660:~/ai-k8s-app$ kubectl get svc
NAME         TYPE        CLUSTER-IP       EXTERNAL-IP   PORT(S)          AGE
ai-service   NodePort    10.101.23.118    <none>        5000:30007/TCP   7m12s
kubernetes   ClusterIP   10.96.0.1        <none>        443/TCP          17m
admin@ubantu-Precision-3660:~/ai-k8s-app$ kubectl get hpa
NAME     REFERENCE           TARGETS       MINPODS   MAXPODS   REPLICAS   AGE
ai-app   Deployment/ai-app   cpu: 3%/50%   2         10        2          5m58s
admin@ubantu-Precision-3660:~/ai-k8s-app$ kubectl top pods
NAME                      CPU(cores)   MEMORY(bytes)
ai-app-5c86c57794-l8hkg   3m           20Mi
ai-app-5c86c57794-mb7rv   3m           20Mi
admin@ubantu-Precision-3660:~/ai-k8s-app$ kubectl describe hpa ai-app
Name:                                                 ai-app
Namespace:                                            default
Labels:                                               <none>
Annotations:                                          <none>
CreationTimestamp:                                    Wed, 11 Mar 2026 16:29:18 +0530
Reference:                                            Deployment/ai-app
Metrics:                                              ( current / target )
  resource cpu on pods  (as a percentage of request):  3% (3m) / 50%
Min replicas:                                         2
Max replicas:                                         10
Deployment pods:                                      2 current / 2 desired
Conditions:
  Type            Status  Reason            Message
  ----            ------  ------            -------
  AbleToScale     True    ReadyForNewScale  recommended size matches current size
  ScalingActive   True    ValidMetricFound  the HPA was able to successfully calculate a replica count from cpu resource utilization (percentage of request)
  ScalingLimited  True    TooFewReplicas    the desired replica count is less than the minimum replica count
Events:           <none>
admin@ubantu-Precision-3660:~/ai-k8s-app$
```